

Tobin Cavanaugh

GitHub: https://github.com/TobinCavanaugh	Email: tobincavanaugh@gmail.com
Personal Site: https://tobincavanaugh.github.io	Phone: +1 206 586 5263

Skills:

<u>Technical Skills</u> • C, C#, Rust, C++, Python, Java • x86-64 asm, Risc-V, SQL, Git, JavaScript, Linux • Debugging	<u>Awards/Certifications</u> • Outstanding CS Student Award, University of Mary • Forklift Certified • Unity Developer Certified	<u>Personal Skills</u> • Out of the box & critical thinking • Strong communicator • Solution oriented
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Work History:

Metrology/Software Dev Internship

Electroimpact	Mukilteo, WA	Summer 2024, 2023, 2025
• Debugged and developed an automation program for the 787 Mid Body Join to be installed at the Boeing Charleston assembly plant. The process involved collaboration with both developers and engineers working on embedded systems. • Engineered a full GUI application for precision controlling and measuring with high-end laser trackers. • Solved laser tracker stand resonance, saving tens of thousands in laser tracker stand replacements. • Installed a Foundation Reference System (FRS) and performed laser tracker accuracy validation. • Successfully validated Leica and FARO laser trackers for Electroimpact and customers, including Boeing, resulting in saving of ten thousand dollars for a new tracker. • Acquired training in Metrolog X4 and robotic arm simulation across numerous training sessions. • Designed robot pathing for 3D scanning objects using the Creaform Metrascan Black Elite mounted on a KUKA robot arm.		

ENR 304 Teachers Assistant

University of Mary	Bismarck, ND	Fall Semester 2025
• Aided students in understanding C programming and provided one-on-one tutoring to improve their Linux fundamentals. • Supported the professor by clarifying lecture material and facilitating problem-solving sessions for students.		

Personal Projects:

sandfleaOS

Personal	Seattle, WA	Fall 2025
• Developed a 64-bit operating system from scratch in C and Assembly, bootable on physical x86 hardware with custom standard library. • Implemented foundational kernel services, including virtual memory management, a task scheduler, and ext2 filesystem support. • Source code currently private		

sstr.h

Personal	Bismarck, ND & Seattle, WA	Summer 2024
• Developed the first C string library to provide immutable string operations returning new string instances, similar to C# string handling but optimized for C • Achieved 5x performance improvement over standard C string functions through custom stack-based memory management and efficient buffer allocation algorithms • Reduced memory-related crashes by 90% by eliminating manual malloc/free management and preventing buffer overflows through automatic memory handling • Implemented custom algorithms using basic data structures (stacks, dynamic arrays) to enable safe string operations like concatenation, substring, and formatting that return new string objects		

Education:

B.Sc. Computer Science	University of Mary	Bismarck, North Dakota	Expected Graduation: April 2027
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